WEEK 1 – DATA SCIENCE AND VISUALIZATION

Project Planning and Dataset Selection
Project: Discovering the Factors Behind Song Popularity Through Data Science

1. Introduction

The first week of the Data Science and Visualization project focused on forming the project foundation, selecting a suitable real-world dataset, understanding the project objective, and establishing the development environment. The project focuses on analyzing Spotify song data to discover meaningful patterns and understand the factors associated with song popularity.

The selected dataset is the Top Spotify Songs 2023 / Spotify Tracks dataset obtained from Kaggle. It contains information about tracks, artists, albums, streaming-related attributes, playlist information, chart-related information, and various audio features. The dataset provides a suitable foundation for applying the complete data science workflow required for the project.

2. Project Title

Discovering the Factors Behind Song Popularity Through Data Science

3. Problem Statement

The global music industry releases a large number of songs every week. Although streaming platforms provide detailed information about songs and their audio characteristics, identifying the factors associated with song popularity remains a challenging task.

This project aims to apply data science and visualization techniques to Spotify track data in order to understand the dataset, assess its quality, perform preprocessing and exploratory analysis, identify relationships between musical features and popularity, and derive meaningful insights from the data.

4. Objectives

The main objectives established during Week 1 are:

• Select and study a real-world Spotify dataset.

• Understand the problem domain and available data.

• Assess the quality and suitability of the selected dataset.

• Identify potential data-cleaning requirements such as missing values, duplicates, inconsistent data types, and outliers.

• Perform exploratory data analysis in the subsequent project stages.

• Investigate relationships between audio features and song popularity.

• Develop meaningful visualizations and insights.

• Prepare the data for advanced analysis, feature engineering, dimensionality reduction, and an interactive dashboard in later stages.

5. Dataset Selection

The project uses a Spotify music dataset obtained from Kaggle. The dataset contains approximately 19,335 song records and multiple attributes related to tracks, artists, albums, and audio characteristics.

Important information available in the dataset includes track name, artist name, album information, popularity, release date, playlist information, chart-related attributes, BPM, musical key, mode, danceability, energy, valence, acousticness, instrumentalness, liveness, speechiness, loudness, tempo, duration, and other Spotify-related fields.

6. Reason for Dataset Selection

The Spotify dataset was selected because it represents real-world music data and contains a combination of numerical and categorical attributes. The audio features provide quantitative variables that can be analyzed statistically, while the track, artist, album, and other metadata provide categorical and textual information.

The dataset also provides opportunities for data-quality assessment, exploratory data analysis, feature transformation, correlation analysis, dimensionality reduction, visualization, and interactive storytelling. These characteristics make it suitable for the Data Science and Visualization project requirements.

7. Initial Data-Quality Considerations

During the initial review, the team identified that the dataset requires detailed quality assessment before analysis. Potential areas for investigation include missing values, duplicate records, data-type consistency, inconsistent values, and numerical outliers.

These issues will be examined systematically in the following weeks. Cleaning decisions will be documented with appropriate reasoning instead of applying transformations without justification.

8. Proposed Methodology

The project will follow an end-to-end data science workflow:

Raw Spotify Dataset

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Data Understanding and Data Quality Assessment

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Data Cleaning

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Exploratory Data Analysis

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Feature Engineering and Transformation

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Feature Selection and Correlation Analysis

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PCA and Dimensionality Reduction

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Interactive Visualization Dashboard

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Insights and Data Storytelling

The methodology will be developed progressively throughout the project timeline.

9. Tools and Technologies

Python will be used as the primary programming language. The main libraries and tools planned for the project are Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn. Jupyter Notebook will be used for analysis and documentation, while GitHub will be used for version control and maintaining weekly project progress. An interactive dashboard will be developed in a later stage using an appropriate dashboarding framework such as Streamlit or Plotly.

10. Team Formation and Responsibilities

The project team consists of four members:

- Prithesa R (25BAD083)
- Saravanan A (25BAD102)
- Tejas T (25BAD124)
- Viknesh A (25BAD128)

Team responsibilities will be distributed across data handling, analysis, visualization, documentation, and presentation activities so that all members contribute to the project development.

11. Week 1 Activities Completed

During Week 1, the team:

• Finalized the Spotify dataset for the project.

• Studied the dataset and its domain.

• Defined the project title and problem statement.

• Identified the main project objectives.

• Prepared the project abstract and dataset presentation.

• Identified the major categories of information available in the dataset.

• Established the initial methodology for the project.

• Set up the GitHub repository for maintaining project files and weekly progress.

12. Expected Outcomes

The project is expected to produce a properly assessed and cleaned Spotify dataset, supported by statistical analysis and visualizations. The analysis will aim to identify meaningful relationships between musical characteristics and popularity and communicate the findings through an interactive dashboard and data-driven story.

The final outcome will include the cleaned dataset, analysis code, visualizations, documentation, and actionable insights derived from the Spotify data.

13. Conclusion

Week 1 established the foundation of the Spotify Data Science and Visualization project. The team selected a suitable real-world dataset, defined the problem and objectives, prepared the initial project documentation, and established the proposed analytical workflow. The next stage will focus on detailed dataset representation, structure inspection, data types, missing-value assessment, descriptive statistics, and the generation of the initial Data Quality/First Impression report.